20/12/2023

Code-A Phase-1

Aakash + BYJU'S

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MM : 720

NCERT Booster Test Series-2023-24_Test-02A

Time : 200 Min.

Topics Covered:
Physics: Alternating Current, Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei
Chemistry: Solutions, Electrochemistry, Chemical Kinetics, Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers
Botany: Ecosystem, Biodiversity and Conservation
Zoology: Biotechnology: Principles and Processes, Biotechnology and Its Applications

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt **all 35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. In sinusoidal alternating current circuit
   (1) The average value of current is zero
   (2) The average value of square of the current is zero
   (3) Average power dissipation is zero
   (4) The phase difference between voltage and current is zero

2. A.C. power is transmitted from a powerhouse at a high voltage as
   (1) Frequency of high voltage is lower
   (2) It is more economical due to less power loss
   (3) Power cannot be transmitted at low voltages
   (4) Both (1) and (2)

3. The alternating e.m.f. of $e = e_0 \sin\omega t$ is applied across a capacitor C. The current through the circuit is given by
   (1) $I = I_0 \sin \omega t$
   (2) $I = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)$
   (3) $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$
   (4) $I = I_0 \sin(\omega t - \pi)$

4. Modified ampere's circuital law holds good for
   (1) Conduction current
   (2) Displacement current
   (3) Both (1) and (2)
   (4) None of these

R → X
C → I leads
V lags

Σ → V leads
Σ lags

5. Microwaves are detected by
   (1) Bolometer
   (2) Point contact diodes
   (3) Thermopiles
   (4) The eye

6. The image formed by a concave mirror is
   (1) Always real
   (2) Always virtual
   (3) Certainly real if the object is virtual
   (4) Certainly virtual if the object is real

7. The equation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ holds true for (symbols have their usual meanings)
   (1) Only concave refracting surface
   (2) Only convex refracting surface
   (3) Both concave and convex refracting surface
   (4) Only planar refracting surface

8. Critical angle of light passing from glass to water is minimum for
   (1) Red colour
   (2) Green colour
   (3) Yellow colour
   (4) Violet colour

9. The image formed by an objective of a compound microscope is
   (1) Real and diminished
   (2) Real and enlarged
   (3) Virtual and enlarged
   (4) Virtual and diminished

10. An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by
   (1) A vernier scale provided on the microscope
   (2) A standard laboratory scale
   (3) A meter scale provided on the microscope
   (4) A screw gauge provided on the microscope

11. In case of reflection over spherical surface, which of these are correct?
   I. Normal is taken as perpendicular of tangent at point of incidence.
   II. Incident ray is always perpendicular to plane of incidence.
   III. Line joining centre of curvature of mirror with point of incidence, can act as normal.
   IV. Line joining centre of curvature and pole of curved surface is always perpendicular to reflected ray.
   (1) I and II
   (2) I and III
   (3) II and III
   (4) II and IV



12. A plane wave passes through a convex lens. The geometrical shape of the wavefront that emerges is
   (1) Plane
   (2) Diverging spherical
   (3) Converging spherical
   (4) None of these

13. If two coherent sources are vibrating in phase, then we have constructive interference at any point P whenever the path difference is [n is integer]
   (1) $\left(n + \frac{1}{2}\right)\lambda$
   (2) $\frac{n\lambda}{2}$
   (3) $\left(n - \frac{1}{2}\right)\lambda$
   (4) $n\lambda$

14. The work-function of a metal is
   (1) The minimum wavelength required to take out electrons from the metal surface
   (2) The maximum frequency required to take out electrons from the metal surface
   (3) The minimum amount of energy required to take out the electrons from the metal surface
   (4) None of these

15. The stopping potential is directly proportional to
   (1) The work function of the metal
   (2) Intensity of incident radiation
   (3) The saturation current for a given frequency
   (4) The kinetic energy gained by the photoelectrons

16. According to Bohr's model of hydrogen atom
(1) The linear velocity of the electron is quantised.
(2) The angular velocity of the electron is quantised.
(3) The linear momentum of the electron is quantised.
(4) The angular momentum of the electron is quantised.

17. Which of the following series in the spectrum of hydrogen atom lies in the visible region of the electromagnetic spectrum?
(1) Paschen series
(2) Balmer series
(3) Lyman series
(4) Brackett series

18. Nucleus of an atom whose atomic mass is 24 can consist of
(1) 11 electrons,11 protons and 13 neutrons
(2) 11 electrons,13 protons and 11 neutrons
(3) 11 protons and 13 neutrons
(4) 11 protons and 13 electrons

19. A transformer is used to light 140 W 24 V lamp from 240 V ac mains. The current in the main cable is 0.7 A. The efficiency of the transformer is
(1) 63.7%
(2) 58%
(3) 83.3%
(4) 34%

20. Which of the following statements are incorrect?
Statement (A): The nuclear force is much stronger than the coulomb force acting between charges or the gravitational force acting between masses.
Statement (B): The nuclear force between two nucleons falls rapidly to zero as their distance is more than a few femtometres. This leads to saturation of forces in a medium or large sized nucleus, which is the reason for the constancy of the binding energy per nucleon.
(1) Only (A)
(2) Only (B)
(3) Both (A) and (B)
(4) Neither (A) nor (B)

21. Apparent depth of a fish 12 m below the surface of water, [refractive index = 4/3] is
(1) 9 m
(2) 19 m
(3) 12 m
(4) 16 m

22. In a series LCR circuit, $R = 10\ \Omega$, $X_C = 14\ \Omega$ and $X_L = 4\ \Omega$. The power factor of the circuit is
(1) $\sqrt{2}$
(2) $\frac{1}{\sqrt{3}}$
(3) $\frac{1}{\sqrt{2}}$
(4) $\frac{1}{2}$

23. The separation between an object (real) and the screen is 100 cm. There are two positions of the convex lens for which image is formed on the screen. The separation between the two positions of the lens is 60 cm. The focal length of the lens is
(1) 10 cm
(2) 13 cm
(3) 16 cm
(4) 18 cm

24. According to Bohr's theory which among the following angular momentum is not possible for a stationary orbit?
(1) $\frac{2h}{3\pi}$
(2) $\frac{3h}{2\pi}$
(3) $\frac{h}{\pi}$
(4) $\frac{5h}{\pi}$

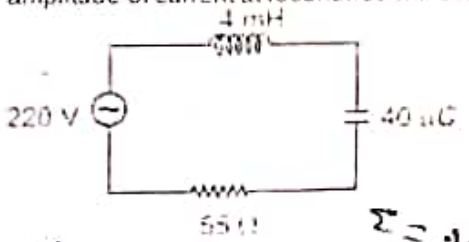
25. An object of 2 cm is placed at a distance of 40 cm in front of a convex lens of focal length 15 cm. The height of the image formed by the lens is
(1) 2 cm
(2) 1 cm
(3) 1.5 cm
(4) 1.2 cm

26. A point source of light is placed at a distance of 2 m from a photocell and cut-off potential is found to be V_0. If distance between point source and photocell is halved, then cut-off potential will be
(1) V_0
(2) $4V_0$
(3) $2V_0$
(4) $\frac{V_0}{2}$

27. If a source of power 4 kW produces 10^{17} photons per second, the radiations belong to a part of the spectrum called
(1) X-rays
(2) Ultra violet rays
(3) Micro waves
(4) Visible rays

28. In YDSE the distance between slits is halved and distance between slit and screen is made two times then width of the fringe
(1) Remains same
(2) Becomes 4 times
(3) Becomes half
(4) Becomes 6 times

29. For the series *LCR* circuit shown in figure, the resonant angular frequency and the amplitude of current at resonance will be



(1) 2500 rad/s and 4 A
(2) 250 rad/s and 4 A
(3) 2500 rad/s and $4\sqrt{2}$ A
(4) 250 rad/s and $2\sqrt{2}$A

30. For a relaxed eye, the magnitude of magnifying power of a compound microscope is (Symbols have their usual meanings)
(1) $\left(\frac{v_0}{u_0}\right)\frac{D}{f_e}$
(2) $\left(\frac{v_0}{u_0}\right)\frac{f_e}{D}$
(3) $\left(\frac{u_0}{v_0}\right)\frac{D}{f_e}$
(4) $\left(\frac{u_0}{v_0}\right)\left(\frac{f_e}{D}\right)$

31. An object 2 cm tall is placed at distance of 5 cm in front of a mirror. In order to produce an upright image of 4 cm height, mirror required should be
(1) Convex mirror of radius of curvature 10 cm
(2) Concave mirror of radius of curvature 10 cm
(3) Concave mirror of radius of curvature 20 cm
(4) Plane mirror of height 10 cm

32. An alternating voltage source V = 260 sin (628t) is connected across a pure inductor of 5 mH. Inductive reactance in the circuit is
(1) 0.318 Ω
(2) 6.28 Ω
(3) 0.5 Ω
(4) 3.14 Ω

33. Transmission of light in optical fibre is due to
(1) Scattering
(2) Diffraction
(3) Refraction
(4) Multiple total internal reflection

34. Refractive index of a medium is given by (symbols have their usual meaings)
(1) $\frac{1}{\sqrt{\mu_r\varepsilon_0}}$
(2) $\sqrt{\mu_0\varepsilon_0}$
(3) $\sqrt{\mu_r\varepsilon_r}$
(4) $\frac{1}{\sqrt{\mu_r\varepsilon_r}}$

35. The ratio of minimum frequency of Paschen series to the minimum frequency of Brackett series will be
(1) $\frac{160}{27}$
(2) $\frac{175}{81}$
(3) $\frac{175}{27}$
(4) $\frac{100}{81}$

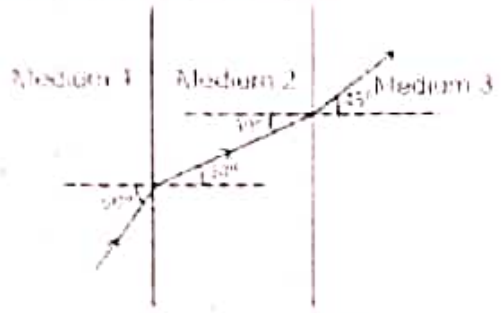
SECTION-B

36. In β minus decay
(1) Z remains unchanged
(2) Z decreases
(3) Z increases
(4) None of these

37. The alternating current can be measured with the help of
(1) Hot wire ammeter
(2) Hot wire voltmeter
(3) Moving magnet galvanometer
(4) Suspended coil type galvanometer

38. The binding energy per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus $_2He^4$ is
(1) 26.3 MeV
(2) 23.6 MeV
(3) 26.3 eV
(4) 23.6 eV

39. The following figure shows refraction of light at the interface of three media



Correct order of optical density(d) of the med ia is
(1) $d_1 > d_2 > d_3$
(2) $d_2 > d_1 > d_3$
(3) $d_1 > d_3 > d_2$
(4) $d_2 > d_3 > d_1$

40. A ray of light is incident normally on one refracting surface of an equilateral prism. If the refractive index of the material of the prism is 1.5, then
(1) The emergent ray is deviated by 30°
(2) The emergent ray is deviated by 60°
(3) The emergent ray just grazes the second refracting surface
(4) The ray undergoes total internal reflection at second refracting surface

41. The focal length of the objective of a telescope is 60 cm. To obtain a magnification of 20 in normal adjustment, the focal length of the eye piece should be
(1) 2 cm
(2) 3 cm
(3) 4 cm
(4) 5 cm

$\frac{60}{20} = \frac{f_o}{f_e}$

3 cm

42. If two sources are coherent, then the phase difference (φ) between the waves produced by them at any point
(1) Will change with time and we will observe stable positions of maxima and minima.
(2) Will not change with time and we will observe unstable positions of maxima and minima.
(3) Will not change with time and we will observe stable positions of maxima and minima.
(4) Will change with time and we will observe unstable positions of maxima and minima.

43. The fringe width in a Young's double slit experiment can be increased if we decrease
(1) Distance between slits and screen
(2) Separation between slits
(3) Wavelength of light used
(4) Both (1) and (3)

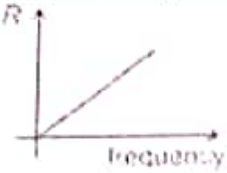
$\beta = \frac{\lambda D}{d}$

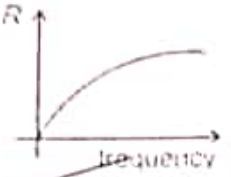
44. The radius of 'n'th Bohr's orbit of H atom is given by
(1) $\frac{\varepsilon_0 n^2 h^2}{\pi m e^2}$
(2) $\frac{n^2 h^2}{\varepsilon_0 \pi m e^2}$
(3) $\frac{\pi m e^2}{\varepsilon_0 n^2 h^2}$
(4) $\frac{\varepsilon_0 h^2}{\varepsilon_0 \pi m e^2}$

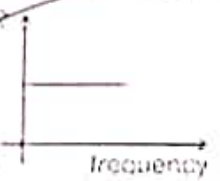
45. Density D of nuclear matter varies with mass number A is
(1) $D \propto A^3$
(2) $D \propto A$
(3) $D \propto A^0$
(4) $D \propto A^2$

46. In Young's double slit experiment, the ratio of path difference corresponding to third maxima obtained in air to the third minima obtained in water is
$\left(\mu \text{ of water } = \frac{4}{3}\right)$
(1) $\frac{4}{3}$
(2) $\frac{4}{5}$
(3) $\frac{3}{5}$
(4) $\frac{8}{5}$

47. Correct variation of resistance in an A.C. resistive circuit as a function of applied a.c. frequency is best represented by the graph

(1) 

(2) 

(3) 

(4) 

48. The nuclides with same neutron number N but different atomic number Z, are called
(1) Isobars
(2) Isotones
(3) Isotopes
(4) Isomers

49. In a series LCR a.c. circuit, the voltage across L, C and R is 20 V each. The voltage of source would be
(1) $20\sqrt{2}$ V
(2) $20\sqrt{3}$ V
(3) $10\sqrt{2}$ V
(4) 20 V

50. The threshold frequency for a certain metal is 2.4×10^{14} Hz. If light of frequency 4×10^{14} Hz is incident on the metal then cut-off voltage of the photoelectric current will be
(1) 0.66 V
(2) 0.99 V
(3) 0.33 V
(4) 1.32 V

CHEMISTRY

SECTION-A

51. The best suited colligative property to determine the molecular mass of polymers is
(1) Elevation of boiling point
(2) Depression of freezing point
(3) Relative lowering of vapour pressure
(4) Osmotic pressure

52. The mole fraction of urea in its 0.1 molal aqueous solution is approximately
(1) $\frac{1}{655.5}$
(2) $\frac{1}{596.5}$
(3) $\frac{1}{556.5}$
(4) $\frac{1}{535.5}$

53. Which among the following concentration terms is independent of temperature?
(1) Molarity
(2) Molality
(3) (w/v)%
(4) Normality

54. Which among the following will show positive deviation from Raoult's law?
(1) $CH_3COCH_3 + CHCl_3$
(2) $C_6H_5OH + C_6H_5NH_2$
(3) $C_2H_5OH + CH_3COCH_3$
(4) $HNO_3 + H_2O$

55. On electrolysis of an aqueous solution of $AgNO_3$ using platinum electrodes, the product obtained at anode will be
(1) H_2
(2) O_2
(3) NO_2
(4) N_2

56. Intercept of graph which is plotted between Λ_m and $c^{1/2}$ for strong electrolyte is (Λ_m plotted on y-axis)
(1) Λ_m^0
(2) $\sqrt{\kappa}$
(3) κ
(4) $\frac{1}{\Lambda_m^0}$



57. The product (B) formed in the following reaction

Cumene $\xrightarrow{O_2/\Delta}$ (A) $\xrightarrow{H^+/H_2O}$ (B) + acetone is
(1) Benzoic acid
(2) Phenol
(3) Toluene
(4) Acetophenone

58. Molar conductivity (in S cm^2 mol^{-1}) of 0.01 M aqueous solution of electrolyte whose cell constant and resistance of solution respectively are 1.2 cm^{-1} and 250 Ω is
(1) 480
(2) 240
(3) 520
(4) 600

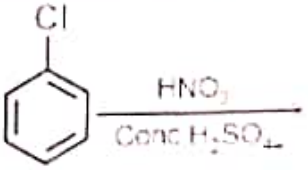
59. Which of the following quantities are affected by the addition of a catalyst during a chemical reaction?
(i) Equilibrium constant change
(ii) Activation energy
(iii) Gibbs energy change
(1) (i) & (ii) only
(2) (ii) & (iii) only
(3) (iii) only
(4) (ii) only

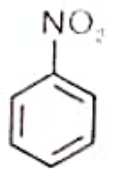
60. Unit of rate constant for a second order reaction is
(1) $mol^{-1} L\ s^{-2}$
(2) $mol\ L^{-1}\ s^{-1}$
(3) s^{-1}
(4) $mol^{-1} L\ s^{-1}$

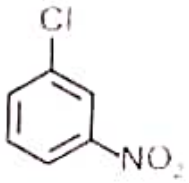
61. The activation energy for a reaction, if its rate triples when the temperature is raised from 30°C to 40°C is
(R = 8.314 J mol^{-1} K^{-1}; log 3 = 0.477)
(1) 10.3 kJ mol^{-1}
(2) 44.3 kJ mol^{-1}
(3) 86.62 kJ mol^{-1}
(4) 160.71 kJ mol^{-1}

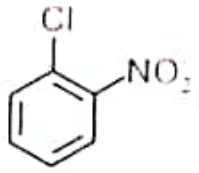
62. In a reaction C + D → product, rate is tripled when the concentration of C is tripled, and the rate increases by a factor of 9 when the concentrations of both the reactants (C and D) are tripled, rate law for the reaction can be written as
(1) Rate = k[C] $[D]^2$
(2) Rate = k[C] [D]
(3) Rate = $k[C]^1$ $[D]^2$
(4) Rate = $k[C]^2$ $[D]^2$

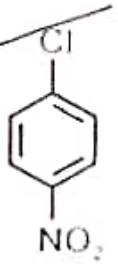
63. Major product of the given reaction is



(1) 

(2) 

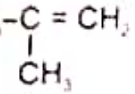
(3) 

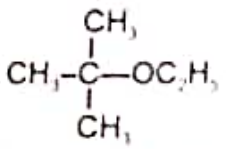
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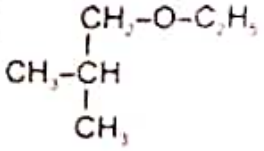
64. In the given reaction, major product formed is



(1) $CH_3-CH=CH-CH_3$

(2) 

(3) 

(4) 

65. Which one is the most reactive towards S_N1 reaction?

(1) C_6H_5Br

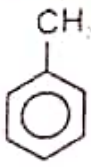
(2) CH_3CH_2Br

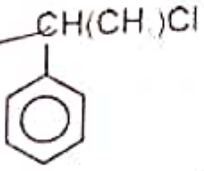
(3) CH_3Br

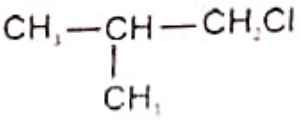
(4) $(CH_3)_3CBr$

66. The compound which will undergo racemisation on reaction with aqueous KOH is

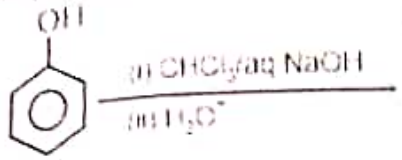
(1) $(CH_3)_3CCl$

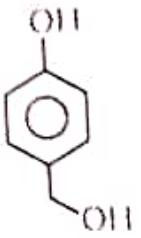
(2) 

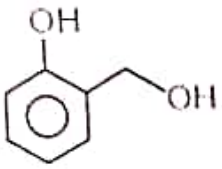
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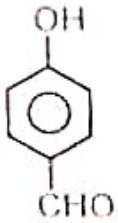
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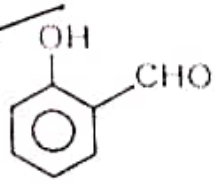
67. Major product of the given reaction is



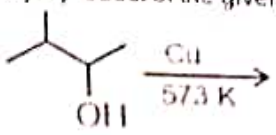
(1) 

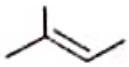
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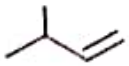
(3) 

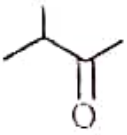
(4) 

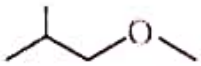
68. Major product of the given reaction is



(1) 

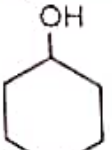
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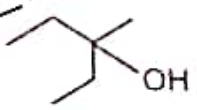
(3) 

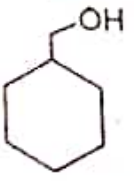
(4) 

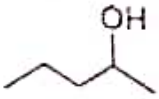
69. Consider the following statements
(a) Ethanol is known as wood spirit
(b) Methanol is more acidic than acetylene
(c) Methanol on reaction with copper at 573 K gives methanal as major product
The correct statements are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a), (b) and (c)
(4) (a) and (c) only

70. The compound which will give immediate turbidity with concentrated HCl and $ZnCl_2$ is

(1) 

(2) 

(3) 

(4) 

71. Which of the following forms azeotropic mixture?
(1) 90% ethanol + 10% water
(2) 75% nitric acid + 25% water
(3) 80% ethanol + 20% water
(4) 68% nitric acid + 32% water

72. The mass of urea required in making 4 kg of 0.4 molal aqueous solution is
(1) 96 g
(2) 93.75 g
(3) 90 g
(4) 100 g

73. Which of the following statement is incorrect with respect to phenomenon of solubility of gas in liquid?
(1) Value of Henry's law constant at 293 K is more than value of Henry's law constant at 303 K for N_2.
(2) The value of Henry's law constant for given gas follows the order Ar > CO_2 > HCHO.
(3) Aquatic species are more comfortable in cold water rather than in warm water because of high solubility of O_2 at low temperature.
(4) Henry's law constant and solubility of gas in liquid are inversely proportional.

74. Consider the following two statements.
Statement I: If the standard electrode potential of an electrode is greater than zero then its reduced form is more stable compared to hydrogen gas.
Statement II: Fluoride ion is the weakest reducing agent.
Choose the correct option from the following.
(1) Statement I and statement II both are correct
(2) Statement I is correct but statement II is wrong
(3) Statement I is wrong but statement II is correct
(4) Both statements I and II are wrong

75. **Assertion:** Λ_m° for weak electrolyte cannot be obtained by extrapolation of Λ_m to zero concentration.
Reason: Λ_m° for weak electrolytes is obtained by using Kohlrausch law of independent migration of ions.
Choose the correct option.
(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
(3) Assertion is correct but Reason is wrong
(4) Both Assertion and Reason are wrong

76. Number of Faradays to obtain 0.5 mole of Cr^{3+} ion from dichromate ion in acidic medium is
(1) 6 F
(2) 3 F
(3) 2 F
(4) 1.5 F

77. The graph of concentration of reactant vs time is straight line for
(1) zero order
(2) First order
(3) Second order
(4) Third order

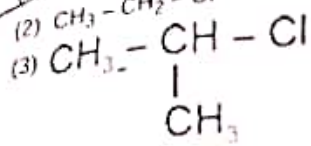
78. For a first order reaction, A → B.
Amount of A is one fourth of its initial concentration after 20 s. The half life of reaction is
(1) 10 s
(2) 40 s
(3) 80 s
(4) 30 s

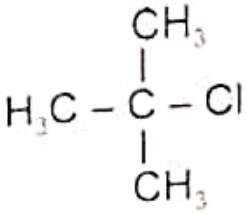
79. The order of the reaction of decomposition of gaseous ammonia on a hot platinum surface at high pressure?
(1) Zero order
(2) First order
(3) Second order
(4) Third order

80. Which of the following undergoes fastest S_N2 reaction?

(1) CH_3-Cl

(2) CH_3-CH_2-Cl

(3) 

(4) 

81. Which of the following hydrocarbons gives single product on monochlorination?

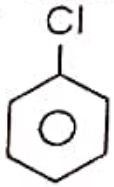
(1) $(CH_3)_4C$

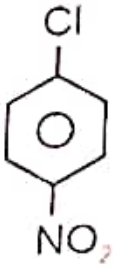
(2) $H_3C-(CH_2)_4-CH_3$

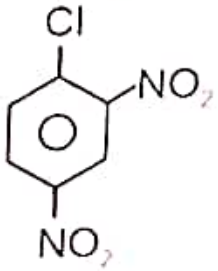
(3) $(H_3C)_2CH-C_2H_5$

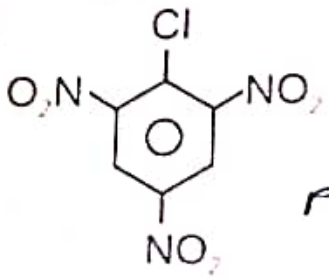
(4) $H_3C-CH_2-CH_2-CH_3$

82. Which among the following will undergo nucleophilic substitution reaction fastest?

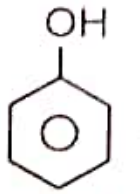
(1) 

(2) 

(3) 

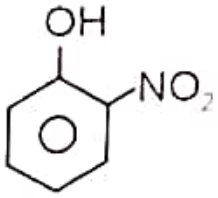
(4) 

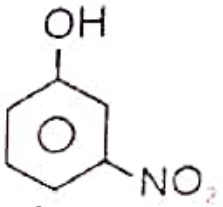
Picric acid

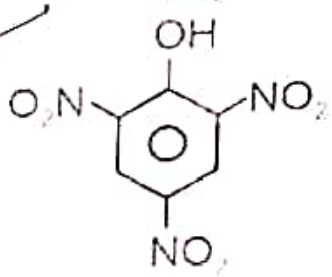
83.  Conc HNO_3 → A

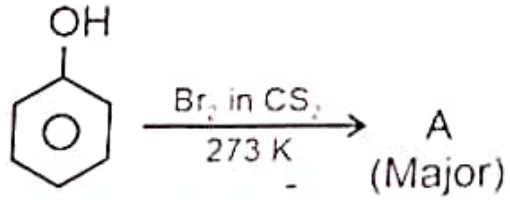
Product A is

(1) 

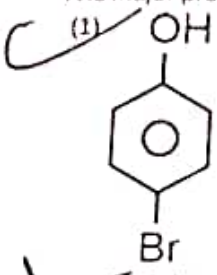
(2) 

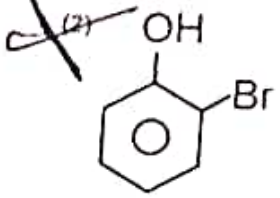
(3) 

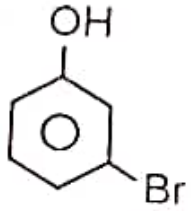
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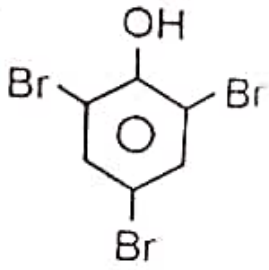
84. 

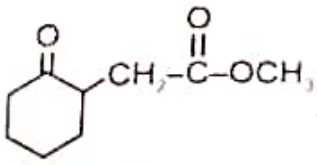
A is major product. Identify A.

(1) 

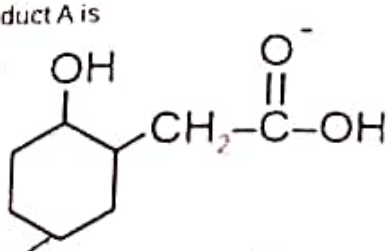
(2) 

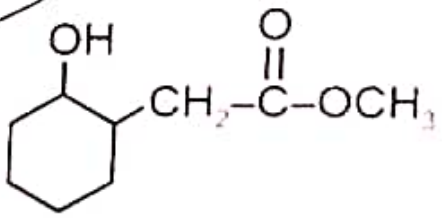
(3) 

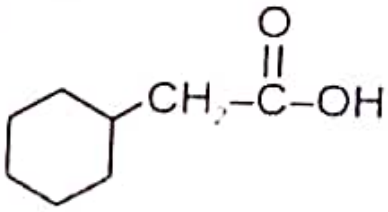
(4) 

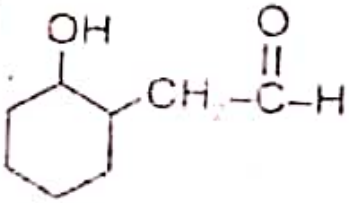
85. 

Product A is

(1) 

(2) 

(3) 

(4) 

SECTION-B

86. The pressure exerted by vapours on the liquid surface in equilibrium with the liquid depends upon
(i) Temperature
(ii) Nature of liquid
(iii) Surface area
(1) (i) only
(2) (ii) & (iii) only
(3) (i), (ii) & (iii)
(4) (i) & (ii) only

87. Which of the following statements is wrong regarding discharging of lead storage battery?
(1) H_2SO_4 is formed in the overall reaction
(2) H_2O is formed in the overall reaction at cathode
(3) $PbO_2(s)$ is reduced to $PbSO_4(s)$ at cathode
(4) $Pb(s)$ is oxidised to $PbSO_4(s)$ at anode

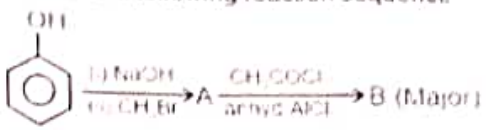
88. If half life of a first order reaction is 10 second then time required for 87.5% completion of reaction is
(1) 15 second
(2) 20 second
(3) 30 second
(4) 40 second

$t_{87.5} = 3 \times t_{1/2}$

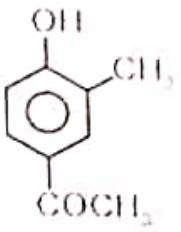
89. Consider the following statements
(a) In S_N1 reaction, carbocation intermediate is formed
(b) S_N1 reactions are generally carried out in polar protic solvents
(c) Greater the stability of carbocation faster is S_N1 rate of reaction
The correct statement are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

$t_{1/2} = \frac{0.693}{k}$

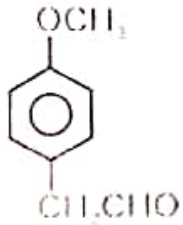
90. Consider the following reaction sequence



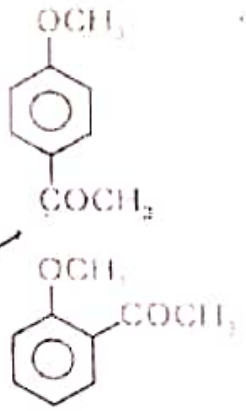
Major product B is
(1)



(2)



(3)



91. Consider the following two statements.
Statement I: Molecules of ethanoic acid dimerise in benzene due to H-bonding.
Statement II: Due to association, the number of particles is reduced.
Select the correct option among the following.
(1) Both statements I and II are correct
(2) Statement I is correct and statement II is false
(3) Statement I is false and statement II is correct
(4) Both statements I and II are false

$t_{87.5}$ $\quad 100 - 87.5 = 12.5$

$\frac{2.303}{kT} \log \frac{12.5}{100} \quad \frac{1}{8}$

92. Consider the following statements.
(a) $CaCl_2$ undergoes 60% dissociation in water. The value of i for $CaCl_2$ is 2.2.
(b) Ethanoic acid undergoes 60% dimerisation in benzene. The value of i is 0.7.
(c) For dissociation i > 1 and for association i < 1.
The correct statements are
(1) (a) and (b)
(2) (b) and (c)
(3) (a) and (c)
(4) (a), (b) and (c)

93. Reduction potential of hydrogen electrode which is placed in solution of pH = 2 is
(1) +0.0591 V
(2) −0.0591 V
(3) −0.1182 V
(4) +0.1182 V

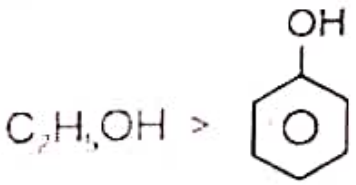
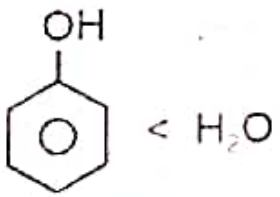
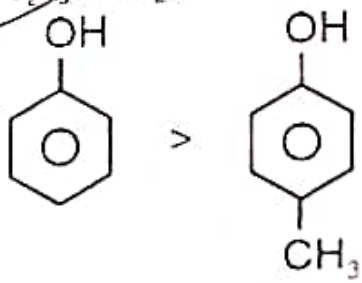
94. For strong electrolytes NaCl, NaOH and $BaCl_2$, the molar conductivities at infinite dilution (in S m^2 mol^{-1}) are x, y and z respectively. The molar conductivity of $Ba(OH)_2$ at infinite dilution (in S m^2 mol^{-1}) will be
(1) z − 2y + 2x
(2) 2z − x + y
(3) z − 2x + 2y
(4) z − 2x − 2y

95. The rate of a reaction get doubled with every 10° rise in temperature. If the temperature of the reaction mixture is raised by 50°C, the new rate of reaction will be
(1) 16 times
(2) 32 times
(3) 64 times
(4) 128 times

96. Rate constant (in mol $L^{-1}s^{-1}$) of a zero order reaction in which reactant concentration changes from 0.3 M to 0.05 M in 10 sec is
(1) 2.5
(2) 0.25
(3) 0.025
(4) 0.0025

97. Which among the following is incorrect statement about freons?
(1) The chlorofluorocarbon compounds of methane and ethane are collectively known as freons.
(2) Freon is non-toxic, non-corrosive and easily liquifiable gas.
(3) Freon is manufactured from tetrachloromethane by Wurtz reaction.
(4) Freons are produced for aerosol propellants, refrigeration and air conditioning purposes.

98. Identify the incorrect statement.
(1) Haloalkanes reacts with KCN to form alkyl cyanides
(2) Haloalkanes reacts with $AgNO_2$ to form alkyl nitrite
(3) Haloalkanes reacts with AgCN to form isocyanides
(4) $^{\ominus}CN$ and NO_2^- are ambident nucleophile

99. Which among the following is correct regarding the acidic strength order of given options?
(1) 
(2) 
(3) $C_2H_5OH > H_2O$
(4) 

100. The correct order of rate of dehydration of following alcohol is
(1) 3° > 2° > 1°
(2) 1° > 2° > 3°
(3) 2° > 1° > 3°
(4) 1° > 3° > 2°

BOTANY

SECTION-A

101. Which of the following statements is correct?
(1) Passenger pigeon become extinct due to co-extinction
(2) Species diversity increases from pole to equator
(3) The species extinction rates during the 'Sixth Extinction' presently in progress, are estimated to be 100 to 1,000 times slower than in the pre-human times
(4) Currently the tropical rain forest occupies 30% of earth land area

102. During succession
(1) Total biomass increases
(2) Types of animals and decomposers do not change
(3) Phytoplankton represents climax community
(4) Changes occur from mesic condition to xeric condition

103. Match the following columns and select the correct option.

Column I	Column II
a. Primary producer	(i) Goat
b. Second trophic level	(ii) Tree
c. Secondary consumer	(iii) Lion
d. Top carnivore	(iv) Wolf

(1) a(ii), b(i), c(iv), d(iii)
(2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(i), c(iv), d(ii)
(4) a(iv), b(i), c(ii), d(iii)

104. Select the **odd** one w.r.t. "The Evil Quartet".
(1) Over-exploitation
(2) Co-extinctions
(3) Reforestation
(4) Alien species invasions

105. What does 'S' stand in the Species-Area relationship equation?
(1) Species area
(2) Species richness
(3) Regression coefficient
(4) Species interrelationship

106. Which of the given taxa contributes maximum to the global biodiversity of vertebrates?
(1) Fishes
(2) Amphibians
(3) Mammals
(4) Birds

107. Hotspots are characterised by all, **except**
(1) Very high level of species richness
(2) Have only rare and threatened organisms
(3) High degree of endemism
(4) Accelerated habitat loss

108. Which of the following is an example of organism with recent extinction in Africa?
(1) Dodo
(2) Quagga
(3) Thylacine
(4) Steller's sea cow

109. Choose the **odd** one for sacred groves.
(1) Khasi and Jaintia hills in Meghalaya
(2) Aravalli hills of Rajasthan
(3) Western Ghat regions of Karnataka and Maharashtra
(4) Bastar area of Tamilnadu

110. A population of *Rauwolfia vomitoria* is adapted to different Himalayan ranges and shows variation in potency and concentration of its active chemical. This way it exhibits
(1) Species diversity
(2) Ecological diversity
(3) Genetic diversity
(4) Community diversity

111. Vertical distribution of different species occupying different levels is called
(1) Standing state
(2) Standing crop
(3) Stratification
(4) Humification

112. Productivity can be expressed in terms of
(1) $gm^{-2}yr^{-1}$
(2) $gm^{-1}yr^{-2}$
(3) $(Kcal\ m^{-1})yr^{-2}$
(4) $(Kcal\ m^{-2})yr^{-1}$

113. The **correct** relationship among net primary productivity, gross primary productivity and respiratory loss is
(1) NPP = GPP – R
(2) NPP = GPP + R
(3) NPP = GPP × R
(4) $NPP = \frac{GPP}{R}$

114. Rate of formation of new organic matter by consumer is called
(1) Gross primary productivity
(2) Net primary productivity
(3) Primary productivity
(4) Secondary productivity

115. Which of the following statements is **incorrect** regarding decomposition of detritus.
(1) Decomposition rate is slower if detritus is in rich in lignin and chitin
(2) Decomposition rate is faster if detritus is rich in nitrogen and water soluble substances
(3) Warm and moist environment favour decomposition
(4) Cold and dry environment favour decomposition

116. In ecological pyramids, the saprophytes
(1) Appear at second trophic level
(2) Appear at third trophic level
(3) Are not given any place
(4) Are included at first trophic level with plants

117. The entire sequence of communities that successively change in a given area is/are called
(1) Seral stages
(2) Sere(s)
(3) Ecological succession
(4) Primary succession

118. One of the examples of areas where secondary succession occurs is
(1) Newly cooled lava
(2) Abandoned farm land
(3) Bare rock
(4) Newly created reservoir

119. Read the given statements and choose the **correct** option:
Statement A: A particular seral stage of succession never gets converted to an earlier stage during primary or secondary succession.
Statement B: Description of ecological succession usually focuses on changes in animals.
(1) Both the statements are true
(2) Both the statements are false
(3) Only statement A is true
(4) Only statement B is true

120. Choose the **odd** one w.r.t. the nutrient cycling in ecosystems.
(1) The standing state remains constant in different kinds of ecosystems but varies on a seasonal basis.
(2) The nutrients are never lost from the ecosystems, rather they are recycled time and again indefinitely.
(3) The reservoir for the sedimentary cycles is located in earth's crust.
(4) The environmental factors, *e.g.*, soil, moisture, pH, temperature, *etc.*, regulate the rate of release of nutrients into the atmosphere.

121. Read the following statements and choose the **correct** option.
A. Among animals, insects are the most species rich taxonomic group.
B. Number of fungal species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.
(1) Both A and B are incorrect
(2) Both A and B are correct
(3) A is correct but B is incorrect
(4) B is correct but A is incorrect

122. Select the **mismatched** pair w.r.t. number of biodiversity conservation strategies in India.
(1) National parks – 90
(2) Hotspots – 3
(3) Biosphere reserve – 14
(4) Wildlife sanctuaries – 148

123. All of the following are alien species in India, **except**
(1) *Parthenium*
(2) *Lantana*
(3) *Eichhornia*
(4) *Triticum aestivum*

Per year

124. Which of the region of India has highest biodiversity?
(1) Eastern Ghats
(2) Western Ghats
(3) Only northern part of Himalaya
(4) Thar Desert

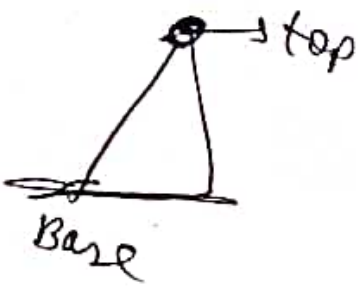
125. How much amount of carbon is fixed annually in the biosphere through photosynthesis?
(1) 4×10^{13} kg
(2) 8×10^{13} kg
(3) 4×10^{17} kg
(4) 4.9×10^{16} kg

126. If 2000 joule of sunlight energy is trapped at producer level, then how much energy will be available to large fish in following food chain.
Phytoplankton → Zooplankton → Small fish → Large fish
(1) 2000 joule
(2) 200 joule
(3) 20 joule
(4) 2 joule

127. The base and apex of each ecological pyramid respectively represent
(1) Producer and tertiary or top level consumer
(2) Tertiary consumer and producer
(3) Secondary consumer and producer
(4) Producer and decomposers

128. Pyramid of energy is always
(1) Upright
(2) Inverted
(3) Spindle shape
(4) Bell shaped

129. The annual net primary productivity of the whole biosphere is approximately
(1) 170 billion tons
(2) 55 billion tons
(3) 110 billion tons
(4) 160 billion tons



130. Select the **incorrect** statement.
(1) A given species may occupy more than one trophic level in the same ecosystem at the same time.
(2) Herbivores obtain phosphorus directly from soil
(3) Ecological pyramid assumes a simple food chain
(4) The gradual and fairly predictable change in the species composition of a given area is called ecological succession

131. Relationship between area and species richness was proposed by
(1) Alexander von Humboldt
(2) Paul Ehrlich
(3) E.P. Odum
(4) David Tilman

132. Consider the given assertion and reason, and choose the **correct** option.
Assertion: The pyramid of biomass in sea is generally inverted.
Reason: The biomass of fishes far exceeds that of phytoplankton.
(1) Both the assertion and reason are true and the reason is the correct explanation of assertion.
(2) Both the assertion and reason are true but the reason is not the correct explanation of assertion.
(3) Assertion is true but the reason is false statement.
(4) Both assertion and reason are false statements.

133. Consider the following statements, state them as **true (T)** or **false (F)** and choose the **correct** option.
A. The species that invade a bare area are called pioneer species.
B. During xeric succession, the lichens pave way to some very small plants like bryophytes, which are able to take hold in the small amount of soil.
C. The climax community remains stable even if the environment changes.

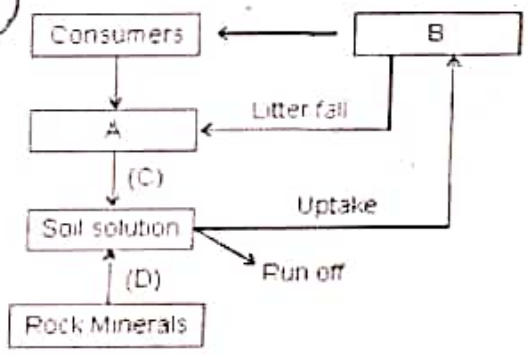
	A	B	C
(1)	T	F	T
(2)	F	T	T
(3)	T	T	F
(4)	T	T	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

134. Out of the total cost of various ecosystem services, the climate regulation accounts for about
(1) 50 per cent
(2) 6 per cent
(3) 10 per cent
(4) 18 per cent

135. All the ecological successions taking place in water and on land, proceeds to
(1) The different climax communities
(2) A similar climax community – the mesic
(3) A similar climax community – the xeric
(4) A similar climax community – the hydric

136.



By looking at the above model of phosphorous cycling in a terrestrial ecosystem identify A, B, C and D respectively.
(1) Detritus, producers, decomposition and weathering
(2) Producers, detritus, decomposition and weathering
(3) Detritus, producers, weathering and decomposition
(4) Consumers, producers, decomposition and weathering

SECTION-B

137. In most ecosystems, all the pyramids of number, energy and biomass are upright. It indicates that
(a) Producers are more in number and biomass than the herbivores.
(b) Herbivores are less in number and biomass than the carnivores.
(c) Energy at a lower tropic level is always more than at a higher level.
Choose the **correct** option.
(1) Only (a) is correct
(2) Only (b) is correct
(3) Both (a) and (c) are correct
(4) Both (b) and (c) are correct

138. Select the **odd** statement w.r.t., detritus food chain (DFC).
(1) DFC begins with dead organic matter
(2) It is made up of decomposers mainly fungi and bacteria
(3) It is a major conduit for energy flow in an aquatic ecosystem
(4) Energy and nutrient requirements are provided by degrading dead organic matter or detritus

139. A secondary consumer in an ecosystem
(1) Is a herbivorous animal
(2) Occupies the second trophic level
(3) Is said to be a primary carnivore
(4) Can occupy the first trophic level

140. Mineralisation
(1) Is the release of simple organic substances from complex inorganic substances
(2) Is the conversion of organic substance from other organic substance
(3) Is the decomposition of detritus to form humus
(4) Is the release of inorganic substances from humus or organic matter

141. Slope of regression (z) for frugivorous birds and mammal in the tropical forests of different continents is found to be
(1) 0.6
(2) 0.1
(3) 1.15
(4) 0.2

142. Match the columns and select the correct option.

	Column I		Column II
A.	Colombia	(i)	More than 1200 species of birds
B.	New York	(ii)	56 species of birds
C.	Greenland	(iii)	105 species of birds
D.	India	(iv)	1400 species of birds

(1) A(iv), B(ii), C(iii), D(i)
(2) A(i), B(iii), C(iv), D(ii)
(3) A(iv), B(iii), C(ii), D(i)
(4) A(iv), B(i), C(iii), D(ii)

143. Who popularised the term 'biodiversity'?
(1) Carolus Linnaeus
(2) Edward Wilson
(3) Ernst Haeckel
(4) Robert May

144. "Lungs of planet" term is used for
(1) Amazon rain forest
(2) Desert
(3) Temperate region
(4) Tundra region

145. Which of the following is **not** a seral community of hydrach succession
(1) Scrub state
(2) Marsh-meadow stage
(3) Zooplankton
(4) Submerged-plant stage

146. The extinction of Steller's sea cow occurred as a result of
(1) Alien species invasion
(2) Co-extinction
(3) Over-exploitation by humans
(4) Habitat loss and fragmentation

147. The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of
(1) African catfish
(2) *Eichhornia*
(3) Cichlid fish
(4) *Clarias gariepinus*

148. "Biodiversity plays a major role in many ecosystem services that nature provides, *e.g.*, pollination."
This argument comes under
(1) The narrowly utilitarian arguments for conserving biodiversity
(2) The broadly utilitarian arguments for conserving biodiversity
(3) The ethical arguments for conserving biodiversity
(4) Bioprospecting

149. Which of the following **do not** serve as *ex situ* conservation of biodiversity?
(1) Zoological parks
(2) Botanical gardens
(3) Wildlife safari parks
(4) Sacred groves

150. The approaches like cryopreservation of the gametes of threatened species and keeping the seeds in seed banks, represent
(1) *In situ* conservation
(2) *Ex situ* conservation
(3) On site conservation
(4) *In situ* and *ex situ* conservation methods respectively

ZOOLOGY

SECTION-A

151. Select the odd one w.r.t. the basic steps in genetically modifying an organism.

(1) Identification of DNA with desirable genes

(2) Introduction of the identified DNA into the host

(3) Separation and purification of genetically modified organism

(4) Maintenance of introduced DNA into the host and transfer of the DNA to its progeny

152. The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme

(1) Restriction endonuclease

(2) DNA ligase

(3) DNA polymerase

(4) *Taq* polymerase

153. In gel electrophoresis, the separated DNA fragments appear as orange coloured bands only after staining with

(1) Calcium chloride

(2) Chilled ethanol

(3) Ethidium bromide

(4) Methylene blue

154. During gel electrophoresis, the DNA fragments move towards A because they are B charged molecules.

Select the option to fill the blanks correctly.

	A	B
(1)	Anode	Negatively
(2)	Anode	Positively
(3)	Cathode	Positively
(4)	Cathode	Negatively

(1) (1)

(2) (2)

(3) (3)

(4) (4)

155. Select the correct recognition sequence for *Eco* RI.

(1) 3'–GAACTT–5'

(2) 5'–GAATTC–3'

(3) 5'–GTTACC–3'

(4) 3'–GTTAAC–5'

156. To make the host cell competent for transformation with rDNA, it is treated with a specific concentration of a

(1) Divalent cation

(2) Monovalent cation

(3) Monovalent anion

(4) Divalent anion

157. When a recombinant DNA is inserted within the coding sequence of β-galactosidase, it results into inactivation of the gene for the synthesis of this enzyme. This is referred to as

(1) Insertional activation

(2) Insertional inactivation

(3) Transformation

(4) Competence

158. Read the statements 'A' and 'B' and choose the **correct** option.

Statement A : *Agrobacterium tumefaciens* causes crown gall in dicots.

Statement B : *Agrobacterium tumefaciens* enters the host through wound and injuries in certain plants.

(1) B is correct, A is incorrect

(2) Both A and B are correct

(3) Both A and B are incorrect

(4) A is correct, B is incorrect

159. Choose the **incorrect** statement w.r.t. cloning vectors.

(1) Plasmids have the ability to replicate within bacterial cells independent of the control of chromosomal DNA.

(2) Disarmed retrovirus is used as vector for cloning of genes of higher organisms.

(3) On linking an alien piece of DNA with bacteriophage, we can multiply its number equal to the copy number of the bacteriophage.

(4) Recombinant DNA molecule can be created by cutting the vector and the source DNA with same DNA ligase.

160. A dsDNA segment is amplified *in vitro* through PCR for 30 cycles. How many copies of the DNA will be obtained at the end of the process?

(1) ~ 30

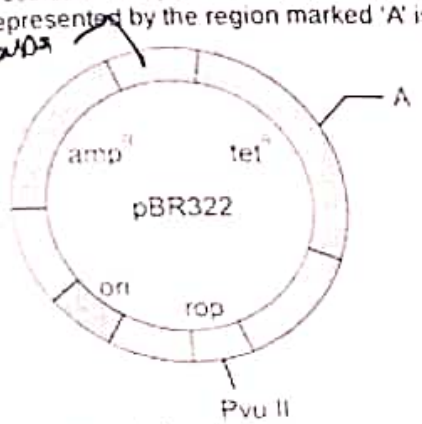
(2) ~ 1 million

(3) ~ 1 billion

(4) ~ 10 billion

161. If gene of interest is cloned at *BamH I* site of pBR322, recombinants will be
(1) Susceptible to tetracycline
(2) Resistant to tetracycline
(3) Susceptible to ampicillin
(4) Not growing on media lacking tetracycline

162. Restriction enzyme whose restriction site is represented by the region marked 'A' is



(1) *Pvu I*
(2) *Bam HI*
(3) *PstI*
(4) *Hind III*

163. Choose the odd one w.r.t palindrome nucleotide sequences.
(1) 5' GAATTC 3'
3' CTTAAG 5'
(2) 5' CCCGGG 3'
3' GGGCCC 5'
(3) 5' CCGC 3'
3' GGCG 5'
(4) 5' GGATCC 3'
3' CCTAGG 5'

164. Match the column I with column II and select the correct option.

	Column I		Column II
a.	Origin of replication	(i)	Process of introducing piece of DNA in a host bacterium
b.	Plasmid	(ii)	Extraction of separated and cut DNA bands from the gel piece
c.	Elution	(iii)	Responsible for initiating replication
d.	Transformation	(iv)	Autonomously replicating circular extra-chromosomal DNA

(1) a(iv), b(iii), c(i), d(ii)
(2) a(iii), b(iv), c(ii), d(i)
(3) a(i), b(ii), c(iv), d(iii)
(4) a(ii), b(i), c(iii), d(iv)

165. Select the correct sequence of steps involved in PCR.
(1) Annealing → Denaturation → Extension
(2) Denaturation → Extension → Annealing
(3) Denaturation → Annealing → Extension
(4) Extension → Annealing → Denaturation

166. All statements are true w.r.t. restriction enzymes except
(1) Restriction enzymes are usually obtained from prokaryotes as well as eukaryotes.
(2) They are the part of natural defense mechanism of prokaryotes against bacteriophages.
(3) They functions by inspecting the length of a DNA sequence.
(4) They can be used to create new combinations of genes.

167. Select the direct method of gene transfer.
(1) Gene transfer using disarmed Ti plasmid of *Agrobacterium*
(2) Gene transfer using disarmed retrovirus
(3) Introduction of gene into host cells using bacteriophages
(4) Introduction of desired gene by micro-injection method

168. Bacteria cell wall degrading enzyme is
(1) Cellulase
(2) Lysozyme
(3) Methylase
(4) Chitinase

169. The novel strategy used by Eli Lilly in 1983 for production of humulin by genetic engineering was
(1) Production of Pro-insulin by *E.coli* and maturation in bioreactors
(2) Production of A and B chains separately and then combined
(3) Production of chains A and B together in the same host cell and then extracted
(4) Production of mature insulin directly by *E.coli* and requiring no further combining of chains

170. Genetic modification in plants have been useful in all of the following ways, **except**
(1) Increased efficiency of mineral usage
(2) Reduced post-harvest losses
(3) Make crops more tolerant to abiotic stresses
(4) Increased reliance on chemical pesticides

171. The applications of biotechnology include all, **except**
(1) Agro-chemical based farming
(2) Production of pest resistant plants
(3) Bioremediation
(4) Therapeutics

172. Transgenic animals are used in toxicity testing of a drug for safety of human beings. In such cases, alien gene is introduced in transgenic animals to make them
(1) More sensitive to toxic substances than non-transgenic animals
(2) Resistant to toxic substances for better tolerance
(3) Less sensitive to toxic substances than normal human beings
(4) Non-reactive to toxic substances for safer use in human beings

173. The method of RNAi involves silencing of a specific A due to formation of B.
Choose the option that fills the blanks **correctly**
(1) **A** - mRNA, **B** - dsRNA
(2) **A** - ssDNA, **B** - dsDNA
(3) **A** - dsDNA, **B** - ssDNA
(4) **A** - dsRNA, **B** - mRNA

174. There are an estimated _____ varieties of rice in India out of which _____ are documented varieties of Basmati. Choose the option which correctly fills the blanks respectively.
(1) 5,00,000; 30
(2) 1,00,000; 50
(3) 2,00,000; 35
(4) 2,00,000; 27

175. Choose the **correct** statement w.r.t probe.
(1) It is always a single stranded RNA, tagged with a radioactive molecule.
(2) It is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography.
(3) It usually have the complementarity with the mutated gene.
(4) It is the single stranded DNA or RNA, not tagged with a radioactive molecule.

176. The first transgenic cow was made to produce human protein-enriched milk in the same year when
(1) The first clinical gene therapy was given
(2) An American company got patent rights on Basmati rice through the US Patent and Trademark Office
(3) An American company prepared Humulin
(4) A four-year old girl got a treatment for adenosine deaminase (ADA) deficiency

177. Given below are some reasons for the production of transgenic animals.
(a) For study of normal physiology and development
(b) Chemical safety and testing
(c) For production of useful biological products
(d) Study of diseases
(e) To test the vaccine safety
How many of the above given reasons are true ?
(1) One
(2) Five
(3) Three
(4) Two

178. Enzyme-Linked Immunosorbent Assay (ELISA) is based on the principle of
(1) Antigen - antibody interaction
(2) Antigen - antigen interaction
(3) Antibody - antibody interaction
(4) Antibody - lipid interaction

179. In case of *BamHI*, '*am*' represents the name of
(1) Genus
(2) Species
(3) Strain
(4) Phylum

180. Read the following statements.
Statement A : The normal *E.coli* does not multiply in the ampicillin containing medium.
Statement B : Transgenic models exist for many human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's.
Choose the **correct** option.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

181. In the case of SCID, which technique is considered to be completely curative?
(1) Oral intake of ADA tablets
(2) Bone marrow transplantation
(3) Enzyme replacement therapy
(4) Gene therapy in early embryonic stage

182. Which one of the following proteins obtained from transgenic animals is being used to treat emphysema?
(1) Insulin
(2) α-1-antitrypsin
(3) Interferon
(4) α-lactalbumin

183. Abnormal gene is replaced by normal gene through
(1) Cloning
(2) Radiation
(3) Gene therapy
(4) Surgery

184. In India, at present how many recombinant therapeutic are being marketed?
(1) 12
(2) 30
(3) 27
(4) 40

185. The full form of EFB is
(1) Eastern Federation of Bioprocess Engineering
(2) Eastern Federation of Bioprocess Biotechnology
(3) European Federation of Biotechnology
(4) European Federation of Bioprocess Engineering

186. How many interchain disulphide bond(s) are present in proinsulin?
(1) One
(2) Two
(3) Three
(4) Zero

SECTION-B

187. Which of the following statements is not **incorrect** w.r.t Bt cotton?
(1) Some strains of *Bacillus thuringiensis* produce proteins that kill certain nematodes
(2) Bt toxin protein exists as active protoxins but once an insect ingest the active toxin, it is converted into an inactive form
(3) The proteins encoded by the genes *cry IAc* and *cry II Ab* control the corn borer and that *cry IAb* controls cotton bollworms
(4) Bt toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides; in effect created a bio-pesticide

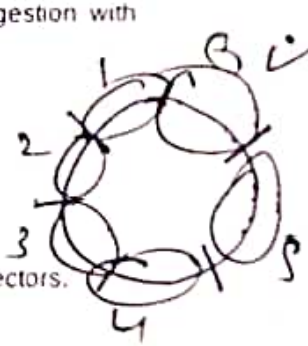
188. In biotechnology, antibiotic resistance genes are used
(1) As selectable markers
(2) As vectors
(3) To maintain sterile condition in culture medium
(4) As enzyme resistant markers

189. In gel electrophoresis,
(1) Smaller fragments travel slower than larger fragments
(2) Larger DNA fragments stay closer to cathode as compared to smaller fragments
(3) Ethidium bromide is used to visualise DNA bands in visible light
(4) DNA is extracted from gel by spooling

190. Downstream processing in biotechnology includes
(1) Introduction of recombinant DNA into host cells
(2) Large scale culture of recombinant bacteria in bioreactors
(3) Isolation of DNA from cell
(4) Separation and purification of recombinant protein

191. Stirrer in stirred-tank bioreactor has been designed for
(1) Providing a temperature control system
(2) Ensuring anaerobic conditions in culture vessel
(3) Addition of preservatives to the product
(4) Facilitating the even mixing and availability of oxygen throughout the process

192. If a circular DNA possesses 6 restriction sites for *BamH*I then how many fragments will be produced upon their digestion with same restriction enzyme?
(1) 6
(2) 7
(3) 8
(4) 9



193. Read the features w.r.t. cloning vectors.
(A) *ori* site
(B) Selectable marker
(C) Cloning site
(D) Recognition site
Select the set which is correct for the features of cloning vectors.
(1) (A) and (B) only
(2) (A), (B) and (C) only
(3) (B) and (C) only
(4) (A), (B), (C) and (D)

194. Agarose is a natural polymer extracted from
(1) Fungus
(2) Bacteria
(3) Sea weeds
(4) Protozoan

195. In the following question, a statement of assertion (A) is followed by a statement of reason (R).
A : Restriction enzymes link foreign DNA to a plasmid in order to produce rDNA.
R : Restriction enzymes are called molecular glue.
In the light of above statements, select the correct option.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

196. In RNAi, host-generated dsRNA triggers protection against nematode infestation as it prevents
(1) Formation of sense-RNA
(2) Translation of a specific mRNA
(3) Formation of antisense-RNA
(4) Lipid synthesis in host plant

197. Bt toxins are
(1) Insect-group specific
(2) Human-group specific
(3) Crop-group specific
(4) Fertiliser-group specific

198. Illegal and unlawful utilisation of biomaterials without payment to people concerned is called
(1) Bio-war
(2) Biopatent
(3) Biopiracy
(4) Biotechnology

199. Select the **correct** statement.

(1) RNAi is a post transcriptional gene silencing method

(2) Golden rice is a drought tolerant variety developed using *Agrobacterium* vector

(3) Human insulin can be produced by bacteria as genetic code is never universal and is species specific

(4) In upstream processing, addition of preservatives in final product plays an essential role

200. Which of the following is **mismatched**?

(1)	Rosie	–	Human protein - enriched milk (2.4 grams per litre)
(2)	Serum analysis	–	Conventional method of disease diagnosis
(3)	PCR	–	Used to check the progression of restriction enzyme digestion
(4)	ELISA	–	Early diagnosis of diseases

(1) (1)

(2) (2)

(3) (3)

(4) (4)